

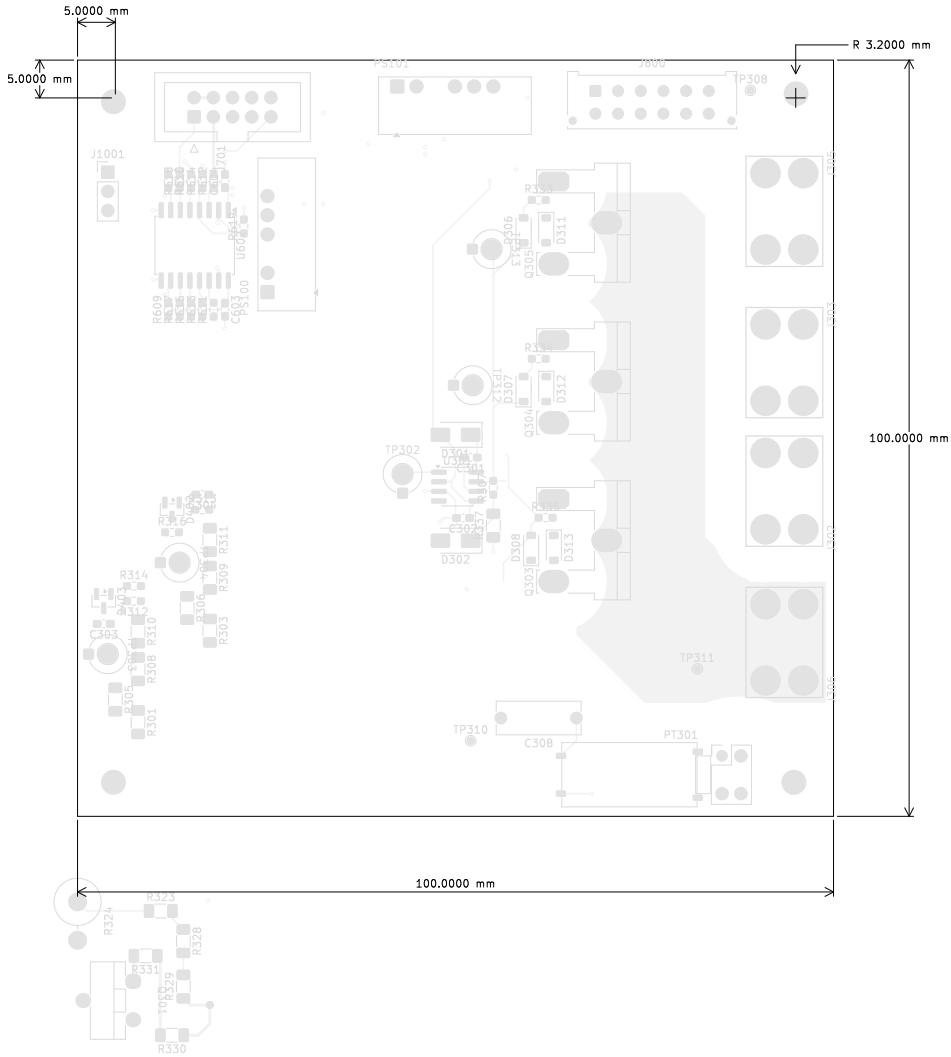
Main board Fabrication Document

Top Fabrication (Scale 1:1)

Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F.Paste				Paste Mask	GBR
F.Silkscreen				Legend	GBR
F.Mask		0.02mm		Solder Resist	GBR
Copper	F.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	In1.Cu (GND)	0.035mm (1.00oz)		Plane	GBR
Core		1.828mm	FR4	Dielectric	
Copper	In2.Cu (Sig, PWR)	0.035mm (1.00oz)		Plane	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	B.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
B.Mask		0.02mm		Solder Resist	GBR
B.Silkscreen				Legend	GBR
B.Paste				Paste Mask	

Total thickness: 2.278mm
Note: external layer thicknesses are specified after plating



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts,GBR" SUFFIX.
DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: HAL SNPB
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:
 - A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - B. Tg 170 C OR EQUIVALENT.
 - C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	100.000 × 100.000 mm
BOARD THICKNESS	2.278 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	1.000 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.300 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.250 mm

HMI
Ext
ctrl

SWD
Status

1 kV strobe
04/Jul/2025

Boot

Light
sense

Felipe L Rodrigues

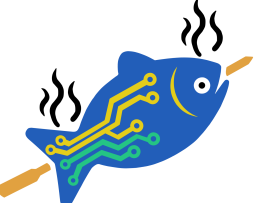
Cap+

Cap-

Lamp
trig

Lamp+

Lamp-

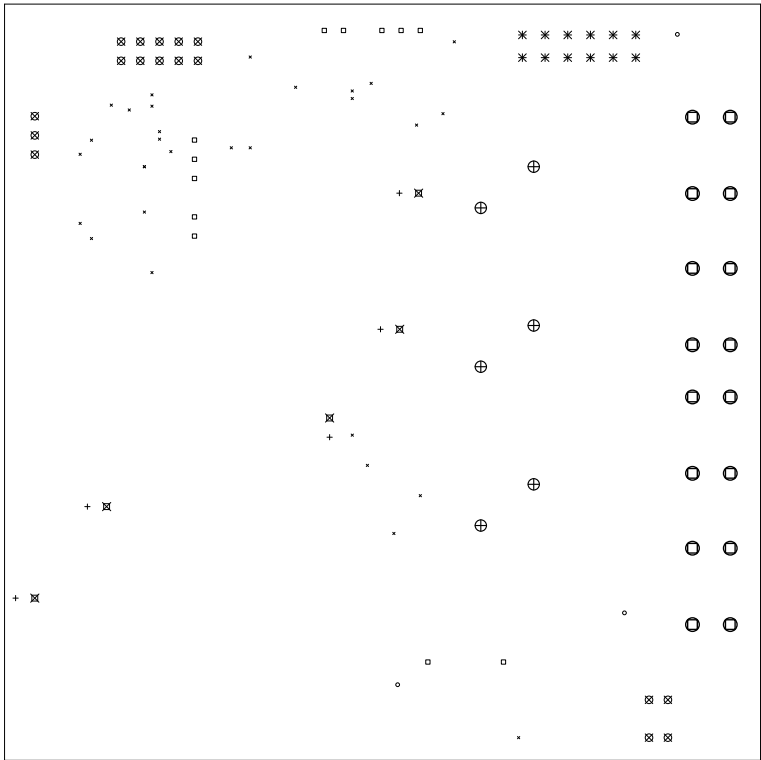
 ChurrasCod Electronics	Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: af0d7df
	Sheet Title: Top Fabrication (Scale 1:1)	Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Path:	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
			Reviewer:	Size: A3	Sheet: 1 of 10

Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	31	0,30mm (11,81mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
○	24	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
+	5	0,70mm (27,56mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
□	12	0,80mm (31,50mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
◇	2	0,90mm (35,43mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊗	20	1,00mm (39,37mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
※	12	1,02mm (40,16mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊠	8	1,10mm (43,31mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊞	2	1,20mm (47,24mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	6	1,50mm (59,06mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊖	16	1,80mm (70,87mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
Total 138						

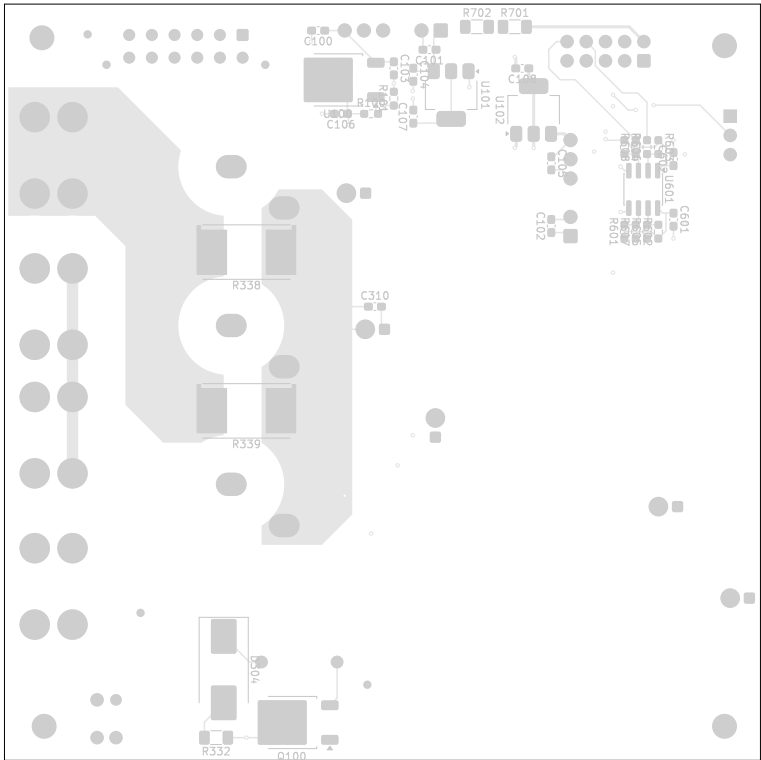


	Comments:		Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: af0d7df
			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)	
	Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 10	

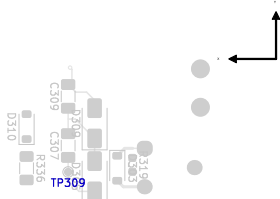
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Main board Fabrication Document

Bottom Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP309	TP_REF	22.50	-19.95
TP502	Net-(U501-PA1)	-32.00	98.00

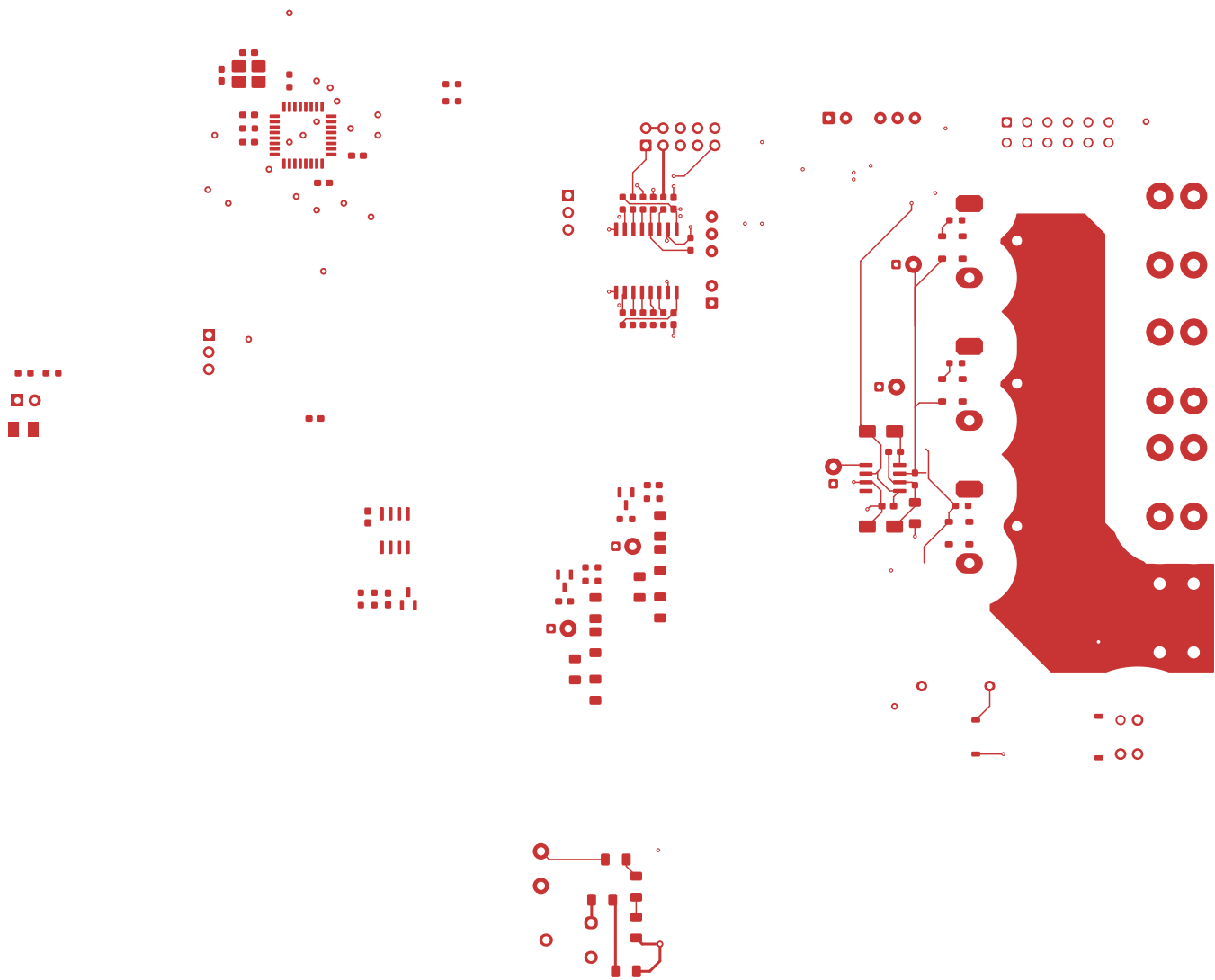


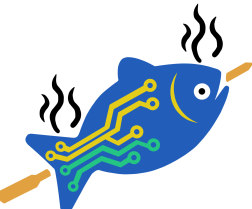
1 kV strobe
04/Jul/25
Felipe L. Rodrigues

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			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)	
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Main board Fabrication Document

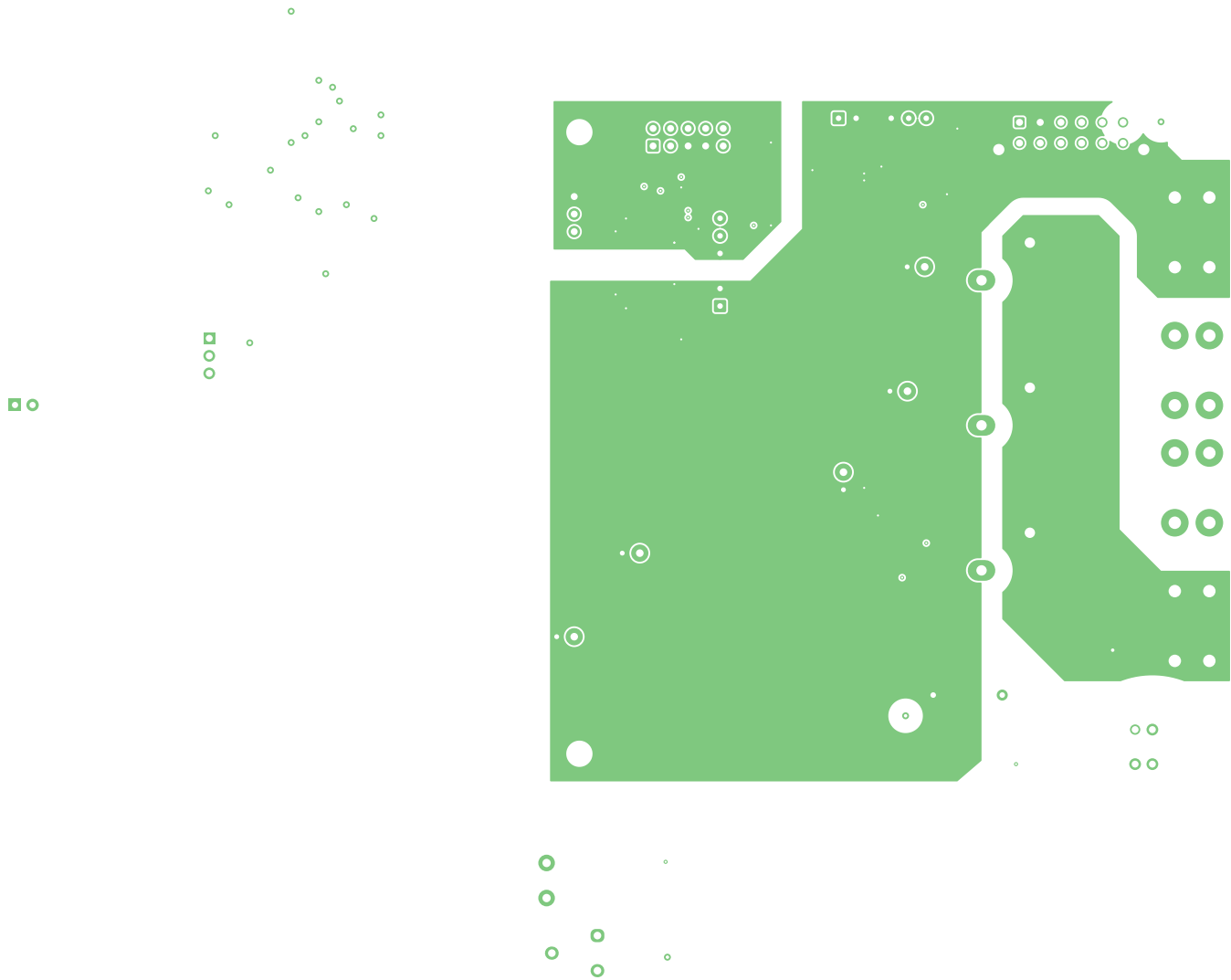
F.Cu (Sig, PWR) (Scale 1:1)

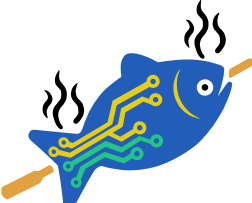


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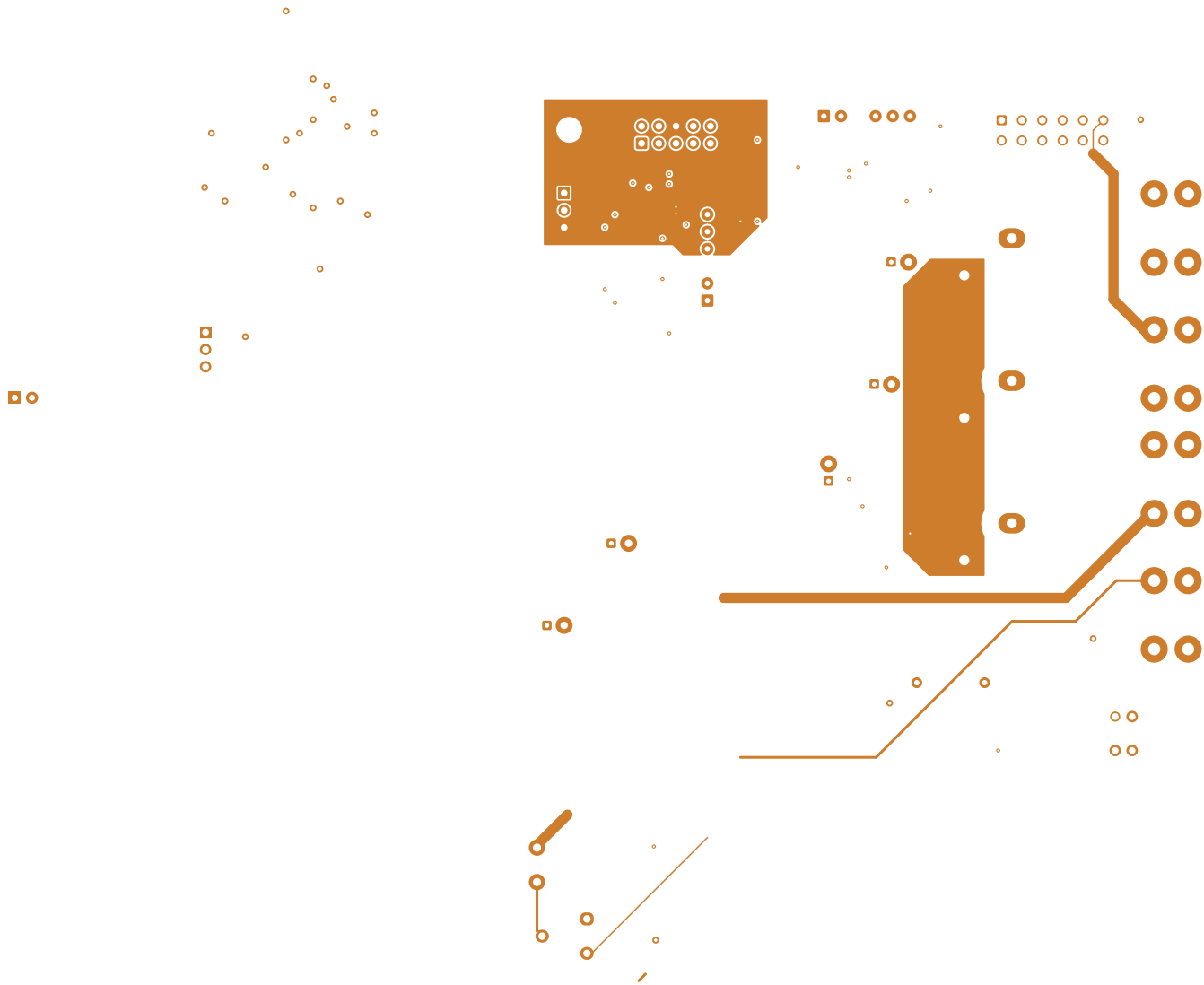
In1.Cu (GND) (Scale 1:1)

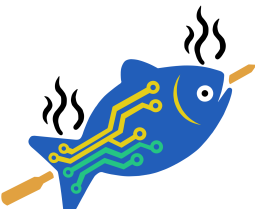


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Main board Fabrication Document

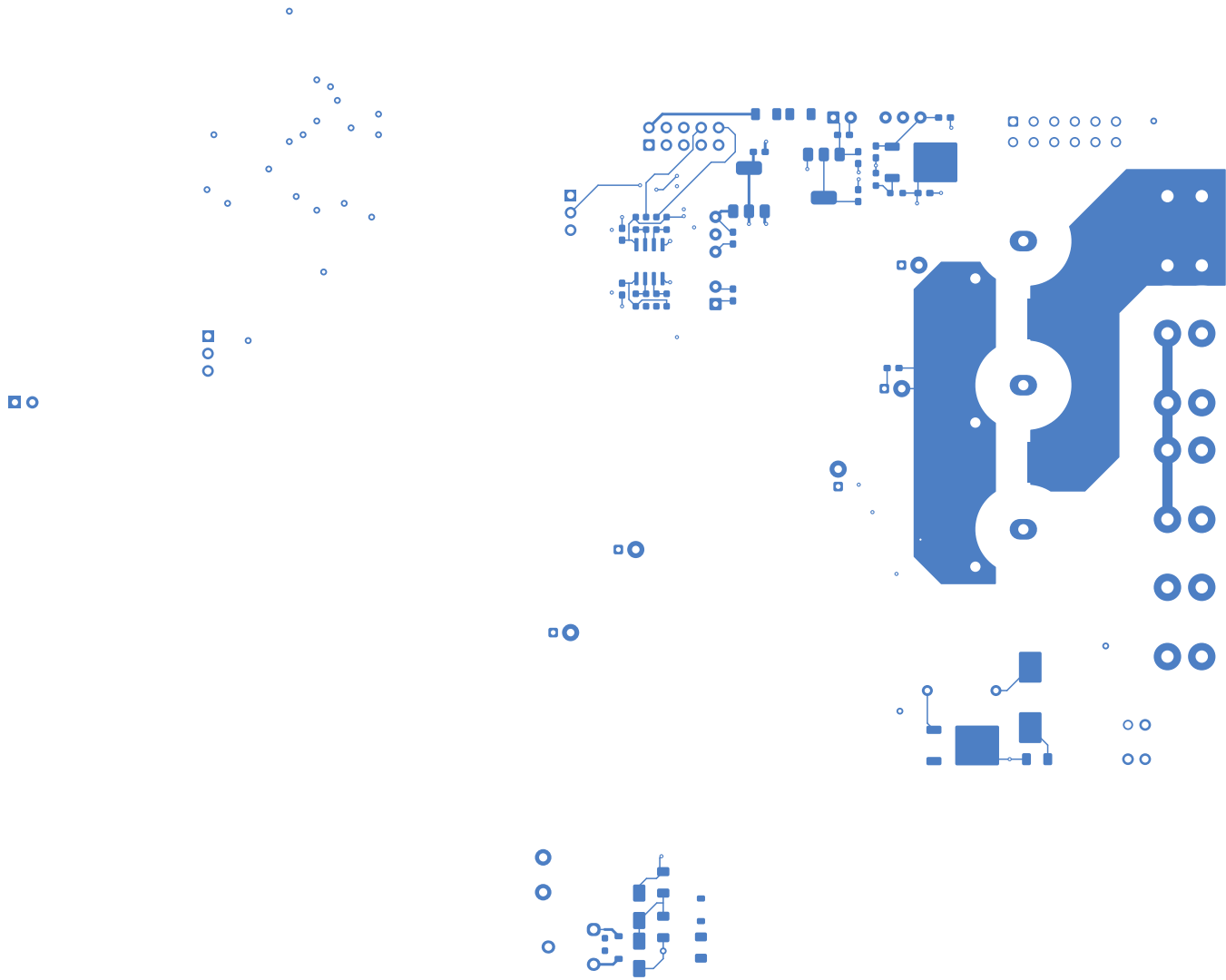
In2.Cu (Sig, PWR) (Scale 1:1)

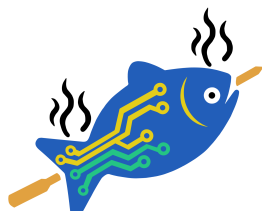


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			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: In2.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb		Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 9 of 10	

Main board Fabrication Document

B.Cu (Sig, PWR) (Scale 1:1)



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			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: B.Cu (Sig, PWR) (Scale 1:1)		File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer:	Size: A3	Sheet: 10 of 10